

Module 3

Chapter 2

Introduction to Gradient

Gradient of a multivariable function

The gradient of a multivariable function $f(x,y,z,\dots)$ is a vector denoted by ∇f that packages all the partial derivatives of the function:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

Let's take an example:

Let's see what ∇f will be for the given function:

$$f(x,y) = x^2 - xy$$

$$\frac{\partial f}{\partial x} = 2x - y$$

$$\frac{\partial f}{\partial y} = -x$$

$$\text{Thus, } \nabla f = \begin{bmatrix} 2x - y \\ -x \end{bmatrix}$$

Let's Practice

Find the gradient of the following function:

$$f(x,y,z) = x - xy + z^2$$

$$f(x, y, z) = x - xy + z^2$$

$$f_x = 1 - y$$

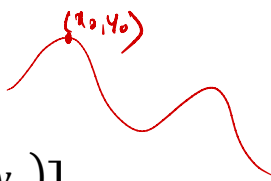
$$f_y = -x$$

$$f_z = 2z$$

$$\therefore \nabla f = \begin{bmatrix} 1 - y \\ -x \\ 2z \end{bmatrix}$$

Interpreting the Gradient

When we calculate the gradient of a function at a given point, say (x_0, y_0) , we essentially convert the input into a vector:


$$\nabla f(x_0, y_0) = \begin{bmatrix} \frac{\partial f(x_0, y_0)}{\partial x} \\ \frac{\partial f(x_0, y_0)}{\partial y} \end{bmatrix}$$

If we think of the graph of f as a hilly terrain and we are standing on the part of the graph directly above—or below—the point (x_0, y_0) , the slope of the hill depends on which direction we walk. For example, if we step straight in the positive x direction, the slope is $\partial f / \partial x$; if we step straight in the positive y -direction, $\partial f / \partial y$. But most directions are some combination of the two.

So, the next question that arises is “*how to find the rate of change of the function if we move in any arbitrary direction?*”

This leads us to our next topic:

Introduction to Directional Derivatives

We know that the *partial derivatives* with respect to x and y tell us the rate of change of f as we nudge the input either in the x or y direction.

The question now is what happens when we nudge the input of f in a direction which is not parallel to the x or y axes.

As you have probably guessed, there is a new type of derivative, called the ***directional derivative***, which answers this question.

Just as the ***partial derivative*** can be taken with respect to some ***input variable***, eg – x or y , the ***directional derivative*** is taken along some vector $\vec{v}$ ***in the input space***.

Notations for Directional Derivatives

The various notations for Directional Derivatives are:

- $\nabla_{\vec{v}} f$
- $\frac{\partial f}{\partial \vec{v}}$
- $f'_{\vec{v}}$
- $D_{\vec{v}} f$
- $\partial_{\vec{v}} f$

All of these represent the same thing: the rate of change of f as you nudge the input along the direction of $\vec{v}$

$$\propto \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Computing the Directional Derivative of a function along a vector

Let's say we have a multi-variable function $f(x,y)$ which takes in 2 variables: x, y ;

We want to compute it's directional derivative along the following vector:

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

We are interested in the **instantaneous rate of change** of the function if we move in the **direction** of the given vector. However, there is a small problem with this.

There can be multiple vectors that point in the same direction. For example, if

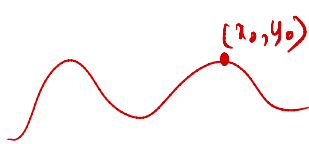
$\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, the vectors $\begin{bmatrix} 6 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 1/5 \\ 1/10 \end{bmatrix}$ etc. all point in the same direction as $\vec{v}$. We

need a way to **consistently** find and define the rate of change of a function in a given direction. We do this by insisting that the vector that defines the direction of change be a unit vector.

So, our **first step** for finding the directional derivative is to find the **unit vector in the direction of $\vec{v}$** . We will call it $\vec{u}$.

$$\vec{u} = \frac{1}{\sqrt{v_1^2 + v_2^2}} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

The **directional derivative** can then be defined as:



$$\nabla f(x_0, y_0) \cdot \vec{u}$$

$$\nabla_{\vec{u}} f = \frac{\partial f}{\partial x} u_1 + \frac{\partial f}{\partial y} u_2 \quad \nabla f(x_0, y_0) \cdot \vec{u}$$

If we are, now, interested in finding the directional derivative of f along $\vec{v}$ at an input point (x_0, y_0) , it will be:

$$\nabla_{\vec{u}} f(x_0, y_0) = \frac{\partial f(x_0, y_0)}{\partial x} u_1 + \frac{\partial f(x_0, y_0)}{\partial y} u_2$$

This can be written as the **dot product** of the **gradient** of the function f at (x_0, y_0) and the unit vector $\vec{u}$:

$$\nabla_{\vec{u}} f(x_0, y_0) = \left[\begin{array}{c} \frac{\partial f(x_0, y_0)}{\partial x} \\ \frac{\partial f(x_0, y_0)}{\partial y} \end{array} \right] \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \nabla f(x_0, y_0) \cdot \vec{u}$$

Let's take an example:

$$\text{Let } f(x, y) = x^2 - xy$$

We want to find the directional derivative of f at point $(2, -3)$ along the vector

$$\vec{v} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

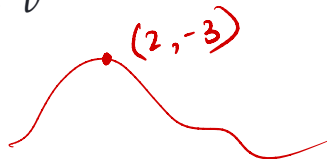
Step 1: Find unit vector $\vec{u}$ in the direction of $\vec{v}$

$$|\vec{v}| = \sqrt{0.6^2 + 0.8^2} = 1$$

So, $\vec{v}$ is a unit vector. So, $\vec{u} = \vec{v}$

Step 2: Find ∇f

$$\nabla f = \begin{bmatrix} 2x - y \\ -x \end{bmatrix}$$



Step 3: Find $\nabla f(2, -3)$

$$\nabla f(2, -3) = \begin{bmatrix} 2 * 2 - (-3) \\ -2 \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$$

Step 4: Find $\nabla_{\vec{u}} f(2, -3)$

$$\nabla_{\vec{u}} f(2, -3) = \nabla f(2, -3) \cdot \vec{u} = \begin{bmatrix} 7 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = 7*(0.6) + (-2)*(0.8) = 2.6$$

*****The most important thing to remember about the gradient:***

The gradient of f , if evaluated at an input (x_0, y_0) , points in the direction of steepest ascent. Why?

We want to **maximize** $\nabla_{\vec{u}} f$ and we know:

$$\nabla_{\vec{u}} f = \nabla f \cdot \vec{u}$$

We need to ask, “which **direction** should we go to **maximize** the above value?” In other words, what **value of $\vec{u}$** will **maximize** the above value?

$$\nabla f \cdot \vec{u} = \underline{|\nabla f| |\vec{u}| \cos \theta}$$

$\theta = 0$
 ∇f and $\vec{u}$ same direction

The length of the vector is fixed (unit vector). So, the only thing to vary is $\cos \theta$, and $\cos \theta$ is **maximum** when θ is 0 ($\cos 0 = 1$). This happens when ∇f and $\vec{u}$ are in the **same direction**.

Thus, we get the **maximum rate of change when we move in the direction of the gradient of the function!**

Let's Practice

1. Find the directional derivative of $f(x, y, z) = x^2 - y^2 + xyz$ in the direction of $\vec{v} = (3, 4, 12)$ at point $(1, 2, -1)$
2. Find the directional derivative of the function $f(x, y, z) = 3xy + z^2$ at the point $(1, -2, 2)$ in the direction from that point towards the origin.
3. Find the maximum rate of change of the given function at point $(-2, 3)$ and give the direction in which this change will occur

$$f(x, y) = \sqrt{x^2 + y^4}$$

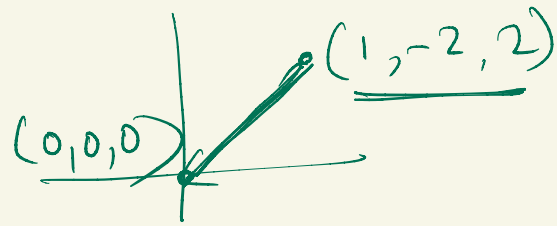
4. The electric potential in a certain region of space is given by the function:

$$V(x, y, z) = 5x^2 - 3xy + xyz$$

- Find the rate of change of voltage at point (3,4,5) in the direction of vector (1, 1, -1)
- In which direction does the voltage change most rapidly at point (3,4,5)

1.

$$\rightarrow \vec{u} = \frac{1}{\sqrt{169}} \begin{bmatrix} 3 \\ 4 \\ 12 \end{bmatrix} = \begin{bmatrix} 3/13 \\ 4/13 \\ 12/13 \end{bmatrix}$$



$$\nabla f(x, y, z) = \begin{bmatrix} 2x + yz \\ -2y + xz \\ xy \end{bmatrix} \rightarrow \text{at } (1, 2, -1) = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$$

$$\nabla_{\vec{u}} f(1, 2, -1) = \begin{bmatrix} 3/13 \\ 4/13 \\ 12/13 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix} = 0 - 20/13 + 24/13 = \boxed{4/13} - \text{Ans}$$

2. $\rightarrow$ Step 1: Figure out $\vec{v}$

$$* \vec{v} \rightarrow \begin{bmatrix} (x_2 - x_1) \\ (y_2 - y_1) \\ (z_2 - z_1) \end{bmatrix}$$

(endpoint - starting point)

$$|\vec{v}| = \sqrt{(-1)^2 + 2^2 + (-2)^2} = \sqrt{9}$$

$$\Rightarrow \vec{v} \rightarrow \begin{bmatrix} (0 - 1) \\ (0 - (-2)) \\ (0 - 2) \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}$$

→ Step 2: Checking if $\vec{v}$ is unit vector. If not, convert to unit vector:

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{1}{\sqrt{9}} \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$$

→ Step 3: Calculate ∇f at $(1, -2, 2)$

$$\nabla f = \begin{bmatrix} 3y \\ 3x \\ 2z \end{bmatrix}$$

$$\nabla f(1, -2, 2) = \begin{bmatrix} 3 \times (-2) \\ 3 \times (1) \\ 2 \times 2 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \\ 4 \end{bmatrix}$$

Step 4: $\nabla_{\vec{u}} f = \nabla f \cdot \vec{u}$

$$= \begin{bmatrix} -6 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1/3 \\ 2/3 \\ -2/3 \end{bmatrix} = \frac{6}{3} + \frac{6}{3} - \frac{8}{3} = \boxed{\frac{4}{3}}$$

$$3. \quad f(x, y) = \sqrt{x^2 + y^4}$$

$$\nabla f = \begin{bmatrix} x(x^2 + y^4)^{-1/2} \\ 2y^3(x^2 + y^4)^{-1/2} \end{bmatrix} = \begin{bmatrix} \frac{x}{\sqrt{x^2 + y^4}} \\ \frac{2y^3}{\sqrt{x^2 + y^4}} \end{bmatrix}$$

$$\nabla f(-2, 3) = \begin{bmatrix} -2/\sqrt{85} \\ 54/\sqrt{85} \end{bmatrix}$$

Max. Rate of change is in the direction of the gradient at that point

$$= \frac{1}{\sqrt{\left(-2/\sqrt{85}\right)^2 + \left(54/\sqrt{85}\right)^2}} \begin{bmatrix} -2/\sqrt{85} \\ 54/\sqrt{85} \end{bmatrix} \cdot \begin{bmatrix} -2/\sqrt{85} \\ 54/\sqrt{85} \end{bmatrix}$$

$$= \frac{\left(\frac{-2}{\sqrt{85}}\right)^2 + \left(\frac{54}{\sqrt{85}}\right)^2}{\sqrt{\left(\frac{-2}{\sqrt{85}}\right)^2 + \left(\frac{54}{\sqrt{85}}\right)^2}}$$

$$= \sqrt{\frac{4}{85} + \frac{2916}{85}} = 5.8611$$

This is nothing but $|\nabla f(-2, 3)|$

$$\rightarrow a). \nabla V = \begin{bmatrix} 10x - 3y + yz \\ -3x + 2z \\ xy \end{bmatrix}$$

$$\nabla V \text{ at } (3, 4, 5) = \begin{bmatrix} 30 - 12 + 20 \\ -9 + 10 \\ 12 \end{bmatrix} = \begin{bmatrix} 38 \\ 6 \\ 12 \end{bmatrix}$$

Rate of change at $(3, 4, 5)$ in the direction of
 $\vec{v} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \nabla_{\vec{u}} V$ where $\vec{u}$ is a unit

vector in the direction of $\vec{v}$

$$\vec{u} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ -1/\sqrt{3} \end{bmatrix}$$

$$\therefore \nabla_{\vec{u}} V = \begin{bmatrix} 38 \\ 6 \\ 12 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ -1/\sqrt{3} \end{bmatrix} = 32/\sqrt{3}$$

b. The voltage changes the most rapidly in the direction of the gradient vector at point $(3, 4, 5)$

$\therefore$ In the direction of $\rightarrow \begin{bmatrix} 38 \\ 6 \\ 12 \end{bmatrix}$